

SIMATS ENGINEERING

ASSESSMENT TOOL 3: Application Based Questions

COURSE NAME:	Cloud Computing and Big Data Analytics	COURSE CODE:	CSA1520
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COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments.* (BL3)

Assessment Tool Weightage: 10%

Dave's Taxonomy: Manipulation (Level 2)
Question Count: 5

Total Marks: 25

Code of Conduct

I **Thamaraiselvi S (192511042)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: **Thamaraiselvi S**

Assessment Tasks

Application Based Questions

1. How would virtualization help a company reduce hardware cost and improve utilization in a new cloud datacenter?
2. A startup needs rapid server provisioning for testing and deployment. Explain how hypervisors and virtual machines support this requirement.
3. How can SOA help integrate cloud services developed by different teams? Explain with an application-oriented example.
4. Explain how identity management and privacy controls should be applied in a federated cloud used by partner organizations.
5. A business wants to migrate from isolated physical servers to a software-defined datacenter. Explain the practical benefits they can expect.

Application Based Questions Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Relevance to Application	25%	Response directly addresses the given use case with strong relevance	Mostly relevant response	Partially relevant	Weak relevance
Use of Cloud Concepts	25%	Applies appropriate concepts accurately	Mostly accurate application	Basic application only	Incorrect concept usage
Practical Insight	20%	Shows strong real-world understanding	Shows reasonable practical understanding	Limited practical insight	No practical insight
Completeness	15%	Response covers all required aspects	Covers most aspects	Covers only basic aspects	Incomplete response
Clarity	15%	Clear and concise answer	Generally clear	Some confusion in expression	Poor clarity

1. How would virtualization help a company reduce hardware cost and improve utilization in a new cloud datacenter?

Virtualization enables multiple virtual machines (VMs) to run on a single physical server using a hypervisor. Instead of purchasing separate servers for every application, the company can consolidate workloads onto fewer physical machines. This significantly reduces hardware procurement costs, power consumption, cooling requirements, and maintenance expenses. Virtualization also improves resource utilization because CPU, memory, storage, and network resources can be shared efficiently among multiple VMs.

Additionally, virtual machines can be created, resized, or migrated easily without interrupting business operations. Features such as live migration, automatic load balancing, and snapshots improve system availability and simplify maintenance. As business demand grows, new VMs can be deployed quickly without investing in additional hardware. Overall, virtualization increases efficiency, reduces operational costs, improves scalability, and maximizes hardware utilization in a cloud datacenter.

2. A startup needs rapid server provisioning for testing and deployment. Explain how hypervisors and virtual machines support this requirement.

Startups often require development, testing, and production environments to be created quickly. Hypervisors allow multiple virtual machines to run on the same physical server, making it possible to provision new servers within minutes. Developers can install different operating systems and software configurations without purchasing new hardware.

Virtual machine templates and snapshots allow teams to clone existing environments instantly, reducing setup time. If testing causes system failures, snapshots enable developers to restore the VM to its previous state quickly. Multiple developers can work independently using separate VMs without affecting each other's work. Virtualization also supports continuous integration and continuous deployment (CI/CD) by enabling automated server creation and deletion during software testing.

As a result, startups benefit from faster software development, reduced infrastructure costs, improved flexibility, efficient testing, and quicker product deployment.

3. How can SOA help integrate cloud services developed by different teams? Explain with an application-oriented example.

Service-Oriented Architecture (SOA) is an architectural approach in which applications are divided into independent services that communicate using standard protocols such as REST APIs or SOAP web services. Each development team can build, update, and maintain its service independently while ensuring seamless communication with other services.

For example, consider an online shopping application. One team develops the customer login service, another develops the product catalog service, another develops the payment gateway, and another develops the delivery tracking system. These services exchange information through APIs. When a customer places an order, the authentication service verifies the user, the inventory service checks product availability, the payment service processes the transaction, and the shipping service updates delivery status.

SOA improves application flexibility, code reusability, scalability, easier maintenance, and faster integration of cloud services developed by different teams.

4. Explain how identity management and privacy controls should be applied in a federated cloud used by partner organizations.

A federated cloud connects multiple cloud providers or partner organizations that share computing resources while maintaining their own infrastructure. Since users from different organizations access shared services, strong identity management and privacy protection are essential.

Identity management should implement Single Sign-On (SSO), allowing users to access multiple cloud services with one login. Multi-Factor Authentication (MFA) adds an extra layer of security by requiring additional verification beyond passwords. Role-Based Access Control (RBAC) ensures users only access resources relevant to their responsibilities.

Privacy controls should include encryption of sensitive data during storage and transmission, secure communication protocols, regular security audits, logging and monitoring of user activities, and compliance with regulations such as GDPR or organizational security

policies. These measures prevent unauthorized access, protect confidential information, ensure accountability, and build trust among partner organizations using the federated cloud.

5. A business wants to migrate from isolated physical servers to a software-defined datacenter. Explain the practical benefits they can expect.

A Software-Defined Datacenter (SDDC) virtualizes all major infrastructure components, including compute, storage, networking, and security. Instead of manually configuring physical hardware, administrators manage resources through centralized software.

Migration to an SDDC provides several practical benefits. Resources can be provisioned automatically within minutes, reducing deployment time for new applications. Workloads can be dynamically allocated based on business demand, improving performance and reducing hardware wastage. Automation reduces manual configuration errors and simplifies infrastructure management.

An SDDC also enhances disaster recovery by supporting automated backups, snapshots, and workload migration between servers. Security policies can be centrally managed across the entire infrastructure. Businesses benefit from improved scalability, reduced operational costs, higher system availability, better resource utilization, faster application deployment, simplified maintenance, and increased business agility.

Overall, migrating to a software-defined datacenter enables organizations to build a modern, flexible, and highly efficient cloud infrastructure capable of supporting future business growth.